www.tntextbooks.in

velocity $\vec{v}$ in $\vec{B}$. As a result, the Lorentz force acts on free electrons in the direction from B to A and is given by the relation

$$\overrightarrow{F_B} = -e\left(\vec{v} \times \vec{B}\right) \qquad (4.4)$$

The action of this Lorentz force is to accumulate the free electrons at the end *A*. This accumulation of free electrons produces a potential difference across the rod which in turn establishes an electric field $\vec{E}$ directed along *BA* (Figure 4.9(b)). Due to the electric field $\vec{E}$, the coulomb force starts acting on the free electrons along *AB* and is given by

$$\overrightarrow{F_E} = -e\vec{E} \qquad (4.5)$$

The magnitude of the electric field $\vec{E}$ keeps on increasing as long as accumulation of electrons at the end *A* continues. The force $\vec{F}_E$ also increases until equilibrium is reached. At equilibrium, the magnetic Lorentz force $\vec{F}_B$ and the coulomb force $\vec{F}_E$ balance each other and no further accumulation of free electrons at the end *A* takes place.

i.e., $$\left|\vec{F}_B\right| = \left|\vec{F}_E\right|$$

$$\left|-e\ (\vec{v} \times \vec{B})\right| = \left|-e\vec{E}\right|$$

$$vB\sin 90° = E$$

$$vB = E \qquad (4.6)$$

The potential difference between two ends of the rod is

$$V = El$$

$$V = vBl$$

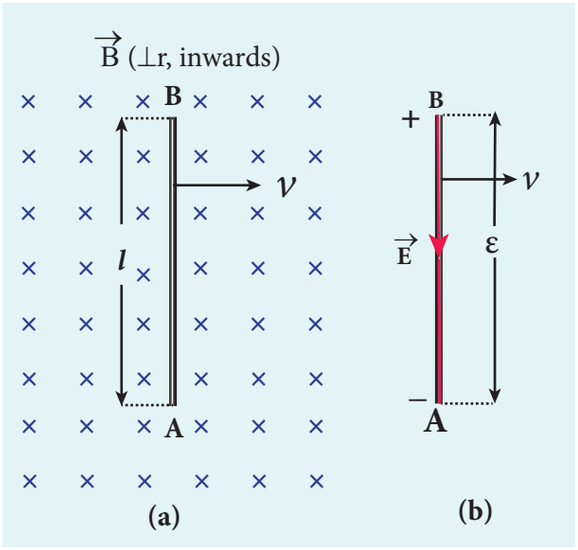


Figure 4.9 Motional emf from Lorentz force

Thus the Lorentz force on the free electrons is responsible to maintain this potential difference and hence produces an emf

$$\varepsilon = Blv \qquad (4.7)$$

As this emf is produced due to the movement of the rod, it is often called as motional emf. If the ends A and B are connected by an external circuit of total resistance *R*, then current $i = \frac{\varepsilon}{R} = \frac{Blv}{R}$ flows in it. The direction of the current is found from right-hand thumb rule.

4.1.7 Motional emf from Faraday's law and Energy conservation

Let us consider a rectangular conducting loop of width l in a uniform magnetic field $\vec{B}$ which is perpendicular to the plane of the loop and is directed inwards. A part of the loop is in the magnetic field while the remaining part is outside the field as shown in Figure 4.10.

UNIT-4(XII-Physics_Vol-1).indd 218 04-03-2019 10:39:06